

# The Math of the Shortprint Application

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This is a high-level overview of the techniques used in the shortprint application for the user who wants to know precisely what is being computed for the sake of accurate inferences. Some statistical background will be assumed.

## Notation

For simplicity, when we say *cards*, we will only be referring to the cards pulled that are of the rarity that the user cares about. We will let  $n$  denote the total number of cards pulled (of that rarity) and  $k$  denote the different number of card names (in that rarity). For simplicity we will label these cards:

$$\{1, 2, 3, \dots, k\} \tag{1}$$

and write  $[k]$  for this set of labels. For  $i \in [k]$ , we will write  $x_i$  for the number of card  $i$  pulled (which we will view as a random variable). We will also write  $p_i$  for the “true” (unknown to us) probability of seeing card  $i$  each time a card of that rarity is pulled. So if there is no shortprinting, then  $p_i = \frac{1}{k}$  for all  $i$ . Finally, we will use the symbol  $\mathbb{P}$  to denote probability.

## Option 1

### Computing the $p$ -value

For this computation only, we will use the normal approximation to the binomial distribution. To calculate the  $p$  value, we assume that all  $p_i = \frac{1}{k}$  and ask how likely it is that our observed ratios  $\{x_i/k\}_{i \in [k]}$  are as different from  $\frac{1}{k}$  as we observe. Under this hypothesis, we have:

$$\sum_{i=1}^k \frac{(x_i - n/k)^2}{n/k} \sim \chi_{k-1}^2 \tag{2}$$

(Pearson's chi-square test); the  $p$ -value we display is accordingly the probability that a  $\chi_{k-1}^2$  random variable is at least as large as the left-hand side of (2).

### The individual confidence intervals

For the individual confidence intervals, we will use the Clopper-Pearson method. Consider the following functions:

$$f^+(p, m) := \sum_{j=m}^n \binom{n}{j} p^j (1-p)^{n-j} \quad (3)$$

$$f^-(p, m) := \sum_{j=0}^m \binom{n}{j} p^j (1-p)^{n-j} \quad (4)$$

for  $p$  a real number in the closed interval  $[0, 1]$  and  $m$  an integer between 0 and  $n$  (inclusive). So for each card  $i$ , and integer  $m$  between 0 and  $n$ , we may express:

$$\mathbb{P}[x_i \geq m] = f^+(p_i, x_i), \quad \mathbb{P}[x_i \leq m] = f^-(p_i, x_i). \quad (5)$$

If we hold  $p$  fixed,  $f^+(p, m)$  is a strictly decreasing function of  $m$ , so

$$\mathbb{P}[f^+(p_i, x_i) \leq f^+(p_i, m)] = \mathbb{P}[x_i \geq m] = f^+(p_i, m). \quad (6)$$

And if  $\beta$  is such that  $f^+(p_i, m) \leq \beta < f^+(p_i, m-1)$ , then we also have

$$\mathbb{P}[f^+(p_i, x_i) \leq \beta] = \mathbb{P}[x_i \geq m] = f^+(p_i, m) \quad (7)$$

(which is at most  $\beta$  by assumption). Altogether, we see that for all  $\beta$  in the closed interval  $[0, 1]$ :

$$\mathbb{P}[f^+(p_i, x_i) \leq \beta] \leq \beta. \quad (8)$$

Using the fact that  $f^-(p, m)$  is strictly increasing in  $m$  (when  $p$  is held fixed), we have

$$\mathbb{P}[f^-(p_i, x_i) \leq f^-(p_i, m)] = \mathbb{P}[x_i \leq m] = f^-(p_i, m) \quad (9)$$

which also the probability that  $f^-(p_i, x_i) \leq \beta$  for  $\beta$  such that  $f^-(p_i, m) \leq \beta < f^-(p_i, m+1)$ . So for all  $\beta$  in the closed interval  $[0, 1]$ :

$$\mathbb{P}[f^-(p_i, x_i) \leq \beta] \leq \beta. \quad (10)$$

Take  $\alpha$  to be the desired confidence (for example,  $\alpha = .95$ ); for each card  $i$ , we have by (8) and (10):

$$\mathbb{P} \left[ f^+(p_i, x_i) \leq \frac{1-\alpha}{2} \text{ or } f^-(p_i, x_i) \leq \frac{1-\alpha}{2} \right] \leq 1-\alpha \quad (11)$$

or in other words

$$\mathbb{P} \left[ f^+(p_i, x_i) > \frac{1-\alpha}{2} \text{ and } f^-(p_i, x_i) > \frac{1-\alpha}{2} \right] \geq \alpha. \quad (12)$$

Combinatorially, one can see that  $f^+(p_i, x_i)$  must be a strictly decreasing function of  $p_i$  and  $f^-(p_i, x_i)$  must be a strictly increasing function of  $p_i$  (holding  $x_i$  fixed and letting  $p_i$  range from 0 to 1). So if we take  $p_{\min}$  to be the (unique) solution to

$$f^-(p_{\min}, x_i) = \frac{1-\alpha}{2} \quad (13)$$

and  $p_{\max}$  to be the (unique) solution to

$$f^+(p_{\max}, x_i) = \frac{1-\alpha}{2} \quad (14)$$

then by (12), we have:

$$\mathbb{P} [p_{\min} < p_i < p_{\max}] \geq \alpha. \quad (15)$$

The confidence interval provided by the application are these values of  $p_{\min}$  and  $p_{\max}$ . Note that  $p_i$  is a fixed and unknown quantity and *not* a random variable (but the  $p_{\min}$  and  $p_{\max}$  are, since they are functions of the random variable  $x_i$ ).

### Simultaneous confidence intervals

Take  $p_{\min,k}$  to be the unique solution to:

$$f^-(p_{\min,k}, x_i) = \frac{1-\alpha}{2k} \quad (16)$$

and  $p_{\max,k}$  to be the unique solution to

$$f^+(p_{\max,k}, x_i) = \frac{1-\alpha}{2k}. \quad (17)$$

For each card  $i$ , the argument of (15) shows:

$$\mathbb{P}[p_{\min,k} < p_i < p_{\max,k}] \geq 1 - \frac{1 - \alpha}{k}; \quad (18)$$

or in other words:

$$\mathbb{P}[p_{\min,k} \geq p_i \text{ or } p_{\max,k} \leq p_i] \leq \frac{1 - \alpha}{k}. \quad (19)$$

Taking the union bound, we have:

$$\mathbb{P}[p_{\min,k} \geq p_i \text{ or } p_{\max,k} \leq p_i \text{ for some } i] \leq 1 - \alpha; \quad (20)$$

in other words

$$\mathbb{P}[p_{\min,k} < p_i < p_{\max,k} \text{ for all } i] \geq \alpha. \quad (21)$$

The interval provided by the application are these values of  $p_{\min,k}$  and  $p_{\max,k}$ .

While it is true that these simultaneous confidence intervals will contain all the true values of  $p_i$  with probability at least  $\alpha$ , our confidence intervals may be slightly narrower than are necessary. But despite the naïveté of the union bound, “cleverer” approaches surprisingly lead to even wider such intervals (see for instance, this paper by Leo Goodman).

## Option 2

### Simultaneous confidence intervals

The simultaneous confidence intervals for Option 2 are computed in the same manner as those for Option 1, except that  $k$  is now smaller.

## Option 3

### Review of Dirichlet distributions

Up until this point, we treated the probabilities  $\{p_i\}_{i \in [k]}$  as fixed unknowns; in Option 3, they are treated as random variables. For notational convenience, we will let:

$$\Delta := \{(p_1, p_2, \dots, p_k) \mid \sum_{i=1}^k p_i = 1 \text{ and all } 0 \leq p_i \leq 1\} \subset \mathbb{R}^k, \quad (22)$$

which is the space of possible values for the  $\{p_i\}$ .  $\Delta$  is a simplex with volume

$$\int_{\vec{x} \in \Delta} 1 \prod_{i=1}^k dp_i = \frac{1}{k!} \quad (23)$$

(several proofs may be found here). Before we discuss the user-inputted prior probability that there is no shortprinting, suppose we were completely agnostic about what the true values of the  $\{p_i\}_{i \in [k]}$  were. The prior distribution on  $\Delta$  would then be:

$$f_{\text{prior}}(p_1, p_2, \dots, p_k) \prod_{i=1}^k dp_i = k! \prod_{i=1}^k dp_i \quad (24)$$

but for simplicity, it is better to view

$$f_{\text{prior}}(p_1, p_2, \dots, p_k) \prod_{i=1}^k dp_i \sim 1 \prod_{i=1}^k dp_i \quad (25)$$

(i.e.  $f_{\text{prior}}(p_1, p_2, \dots, p_k)$  is some constant multiple of 1), and not worry about the exact constant of proportionality. How should we update after making our observations of the  $\{x_i\}_{i \in [k]}$ ? Conditional on given values of the  $\{p_i\}$ , the probability of our observations are:

$$\prod_{i=1}^k (p_i)^{x_i} \quad (26)$$

so via Bayes' theorem, our posterior distribution is:

$$f_{\text{post}}(p_1, p_2, \dots, p_k) \prod_{i=1}^k dp_i \sim \prod_{i=1}^k (p_i)^{x_i} \prod_{i=1}^k dp_i. \quad (27)$$

What is the constant of proportionality? From Wikipedia (see also this stackexchange explanation):

$$\int_{\Delta} \prod_{i=1}^k (p_i)^{x_i} \prod_{i=1}^k dp_i = B(x_1 + 1, x_2 + 1, \dots, x_k + 1) \quad (28)$$

(where  $B$  is the multivariate Beta function, described here in terms of the Gamma function). Since the  $\{x_i\}$  are all non-negative integers, we have:

$$B(x_1 + 1, x_2 + 1, \dots, x_k + 1) = \frac{(x_1)!(x_2)! \cdots x_k!}{n!} = 1 / \binom{n}{x_1, x_2, \dots, x_k}, \quad (29)$$

using the fact that  $\Gamma(x) = (x-1)!$  for positive integers  $x$ . We may then express our posterior as:

$$f_{\text{post}}(p_1, p_2, \dots, p_k) \prod_{i=1}^k = \binom{n}{x_1, x_2, \dots, x_k} \prod_{i=1}^k (p_i)^{x_i} \prod_{i=1}^k dp_i. \quad (30)$$

These posteriors are known as *Dirichlet* distributions. When new observations come in, one simply updates the  $\{x_i\}$  and recomputes the above distribution.

### Averaging a discrete and continuous prior

We will take  $q$  to be the user-inputted prior probability that there is no shortprinting. So with probability  $q$ , we have

$$p_i = \frac{1}{k} \text{ for all } i \in [k]. \quad (31)$$

To simplify the application, we have assumed that the prior on the rest of  $\Delta$  is uniform, but we will discuss in the next subsection how one can modify this to a more general prior. Our prior is now:

$$q\delta_{\frac{1}{k}} + (1-q) \prod_{i=1}^k dp_i \quad (32)$$

where  $\delta_{\frac{1}{k}}$  is the Dirac measure with support at the point  $(\frac{1}{k}, \frac{1}{k}, \dots, \frac{1}{k}) \in \mathbb{R}^k$ . For the reader unfamiliar with measure theory,  $\delta_{\frac{1}{k}}$  may be viewed as a probability distribution that is very, very narrowly concentrated at the point  $(\frac{1}{k}, \frac{1}{k}, \dots, \frac{1}{k})$ , but still has integral 1 (informally speaking, this distribution is “infinitely narrow” and “infinitely concentrated” at that point). We did not write  $f_{\text{prior}}$  because the Dirac measure *cannot* be expressed as the product of a measurable function and the uniform distribution  $\prod_{i=1}^k dp_i$ .

Updating is not too different than the previous case. The probability of our outcomes conditional on all  $p_i = \frac{1}{k}$  is

$$\prod_{i=1}^k \frac{1}{k^{x_i}} = \frac{1}{k^n} \quad (33)$$

so directly applying Bayes’ theorem, the posterior must be proportional to

$$\frac{q}{k^n} \delta_{\frac{1}{k}} + (1-q) \prod_{i=1}^k (p_i)^{x_i} dp_i. \quad (34)$$

Integrating the above over  $\Delta$  gives  $\frac{q}{k^n} + (1 - q)/\binom{n}{x_1, x_2, \dots, x_k}$  (using (28)) so our posterior distribution is

$$\frac{\frac{q}{k^n} \delta_{\frac{1}{k}} + (1 - q) \prod_{i=1}^k (p_i)^{x_i} dp_i}{\frac{q}{k^n} + (1 - q)/\binom{n}{x_1, x_2, \dots, x_k}}. \quad (35)$$

### More general priors (not included in application)

It might feel unnatural to use have the prior be uniform outside of the point  $(\frac{1}{k}, \frac{1}{k}, \dots, \frac{1}{k})$ . If you knew shortprints would occur, you would probably bet on specific cards being printed less than others. As long as one restricts oneself to priors of the form:

$$q \delta_{\frac{1}{k}} + (1 - q) \prod_{i=1}^k \frac{\prod_{i=1}^k (p_i)^{\alpha_i} dp_i}{B(\alpha_1 + 1, \alpha_2 + 1, \dots, \alpha_k + 1)}, \quad (36)$$

(where the  $\{\alpha_i\}_{i \in [k]}$  are non-negative real numbers) this is still quite easy to work with. To pick the  $\{\alpha_i\}_{i \in [k]}$ , consider the following thought experiment: Suppose you knew for a fact that there would be shortprints in this set but also forgot everything you saw about the data. What would you expect to see in the first 100 cards pulled from that rarity? For instance, you could look at all past instances of shortprints and use the ratios that seem to appear the most often. For each  $i \in [k]$ , take  $\alpha_i$  to be your guess for the number of card  $i$  that would appear in the first 100 pulls of that rarity. (The number 100 can of course be replaced with any convenient quantity.) If choosing the  $\alpha_i$  in this manner, the Beta function may be rewritten via (29) since the arguments would all be positive integers. Analogously to (34), the posterior distribution would be proportional to

$$\frac{q}{k^n} \delta_{\frac{1}{k}} + \frac{(1 - q) \prod_{i=1}^k (p_i)^{x_i + \alpha_i} dp_i}{B(\alpha_1 + 1, \alpha_2 + 1, \dots, \alpha_k + 1)} \quad (37)$$

and the integral over this would be

$$\frac{q}{k^n} + \frac{(1 - q) B(\alpha_1 + x_1 + 1, \alpha_2 + x_2 + 1, \dots, \alpha_k + x_k + 1)}{B(\alpha_1 + 1, \alpha_2 + 1, \dots, \alpha_k + 1)}. \quad (38)$$

The posterior distribution would be (37) divided by (38). If the  $\{\alpha_i\}$  were all taken to be integers that summed to 100, we could write:

$$\frac{B(\alpha_1 + x_1 + 1, \dots, \alpha_k + x_k + 1)}{B(\alpha_1 + 1, \dots, \alpha_k + 1)} = \frac{\binom{n}{\alpha_1, \dots, \alpha_k}}{\binom{n+100}{\alpha_1+x_1, \dots, \alpha_k+x_k}} \quad (39)$$

$$= \frac{n!}{(100+n)!} \prod_{i=1}^k \frac{(\alpha_i + x_i)!}{(x_i)!}. \quad (40)$$

We did not include this flexibility in the application as the explanation for its use would be too technical for a general audience. But the user may copy and suitably modify the code of the application to serve this purpose if they wish.

### Credible intervals

Finally we come to what the application actually computes in Option 3: the credible intervals for the desired confidence, which we will call  $\alpha$  (and we encourage the reader to set  $\alpha = .95$  when reading the subsequent discussion). To do so, we begin by computing the marginal distribution of each  $p_i$  in the joint posterior distribution in (35). We will slightly rewrite this joint posterior as:

$$\frac{\frac{q}{k^n} \delta_{\frac{1}{k}} + \frac{(1-q)}{\binom{n}{x_1, x_2, \dots, x_k}} \binom{n}{x_1, x_2, \dots, x_k} \prod_{i=1}^k (p_i)^{x_i} dp_i}{\frac{q}{k^n} + (1-q) / \binom{n}{x_1, x_2, \dots, x_k}}, \quad (41)$$

to view it as a weighted average of the Dirac distribution  $\delta_{\frac{1}{k}}$  and the Dirichlet distribution  $\binom{n}{x_1, x_2, \dots, x_k} \prod_{i=1}^k (p_i)^{x_i} dp_i$ . The marginal distribution of  $\delta_{\frac{1}{k}}$  on  $p_i$  is itself a Dirac distribution with support  $\frac{1}{k}$  (which we'll denote by  $\delta_{\frac{1}{k}, p_i}$ ). And the marginal distribution of the Dirichlet distribution  $\binom{n}{x_1, x_2, \dots, x_k} \prod_{i=1}^k (p_i)^{x_i} dp_i$  is itself the Dirichlet distribution:

$$\binom{n}{x_i} (p_i)^{x_i} (1 - p_i)^{n-x_i} dp_i \quad (42)$$

(see again Wikipedia). Taking the appropriate weighted averages, the marginal posterior on each  $p_i$  is:

$$\frac{\frac{q}{k^n} \delta_{\frac{1}{k}, p_i} + \frac{(1-q)}{\binom{n}{x_1, x_2, \dots, x_k}} \binom{n}{x_i} (p_i)^{x_i} (1-p_i)^{n-x_i} dp_i}{\frac{q}{k^n} + (1-q)/\binom{n}{x_1, x_2, \dots, x_k}}. \quad (43)$$

Let now  $\alpha$  be the user-inputted desired credible level (for instance,  $\alpha = .95$ ). We want to find a “good” choice of  $\ell_i$  and  $r_i$  such that  $0 \leq \ell_i \leq r_i \leq 1$  and:

$$\int_{p_i=\ell_i}^{r_i} \left( \frac{\frac{q}{k^n} \delta_{\frac{1}{k}, p_i} + \frac{(1-q)}{\binom{n}{x_1, x_2, \dots, x_k}} \binom{n}{x_i} (p_i)^{x_i} (1-p_i)^{n-x_i} dp_i}{\frac{q}{k^n} + (1-q)/\binom{n}{x_1, x_2, \dots, x_k}} \right) \geq \alpha; \quad (44)$$

in other words,  $\mathbb{P}[\ell_i \leq p_i \leq r_i] \geq \alpha$ . A naïve approach might be to solve:

$$\mathbb{P}[p_i \leq \ell_i] = \frac{1-\alpha}{2}, \quad \mathbb{P}[p_i \geq r_i] = \frac{1-\alpha}{2} \quad (45)$$

for  $\ell_i$  and  $r_i$  but the danger is that the Dirac distribution might prevent solutions from occurring! Instead, we find the *maximum*  $\ell_i$  and *minimum*  $r_i$  such that:

$$\mathbb{P}[p_i < \ell_i] \leq \frac{1-\alpha}{2}, \quad \mathbb{P}[p_i > r_i] \geq 1 - \frac{1-\alpha}{2}. \quad (46)$$

For notational simplicity, we will set

$$A := \left( \frac{q}{k^n} + (1-q)/\binom{n}{x_1, x_2, \dots, x_k} \right) \frac{1-\alpha}{2} \quad (47)$$

so we may write these requirements as:

$$\int_{p_i=0}^{p_i=\ell_i} \frac{q}{k^n} \delta_{\frac{1}{k}, p_i} + \frac{(1-q)}{\binom{n}{x_1, x_2, \dots, x_k}} \binom{n}{x_i} (p_i)^{x_i} (1-p_i)^{n-x_i} dp_i \geq A \quad (48)$$

$$\int_{p_i=r_i}^{p_i=1} \frac{q}{k^n} \delta_{\frac{1}{k}, p_i} + \frac{(1-q)}{\binom{n}{x_1, x_2, \dots, x_k}} \binom{n}{x_i} (p_i)^{x_i} (1-p_i)^{n-x_i} dp_i \geq A. \quad (49)$$

We begin with  $\ell_i$ , and calculate:

$$L := \int_{p_i=0}^{p_i=1/k} \frac{(1-q)}{\binom{n}{x_1, x_2, \dots, x_k}} \binom{n}{x_i} (p_i)^{x_i} (1-p_i)^{n-x_i} dp_i. \quad (50)$$

If  $L \geq A$ , then our desired value of  $\ell_i$  is the solution to:

$$\int_{p_i=0}^{p_i=\ell_i} \frac{(1-q)}{\binom{n}{x_1, x_2, \dots, x_k}} \binom{n}{x_i} (p_i)^{x_i} (1-p_i)^{n-x_i} dp_i = A \quad (51)$$

(which will be at most  $\frac{1}{k}$ ). Rewriting this as:

$$\int_{p_i=0}^{p_i=\ell_i} \binom{n}{x_i} (p_i)^{x_i} (1-p_i)^{n-x_i} dp_i = \frac{A \binom{n}{x_1, x_2, \dots, x_k}}{(1-q)}, \quad (52)$$

this value of  $\ell_i$  may be obtained using the `scipy.stats.beta.ppf` function in Python's Scipy package. If on the other hand,  $L < A$ , we next check if

$$L + \frac{q}{k^n} \geq A. \quad (53)$$

If it is, then our desired value of  $\ell_i$  is  $\frac{1}{k}$ . If we have  $L + \frac{q}{k^n} < A$ , then  $\ell_i$  is the solution to:

$$\frac{q}{k^n} + \int_{p_i=0}^{p_i=\ell_i} \frac{(1-q)}{\binom{n}{x_1, x_2, \dots, x_k}} \binom{n}{x_i} (p_i)^{x_i} (1-p_i)^{n-x_i} dp_i = A \quad (54)$$

(which will be strictly greater than  $\frac{1}{k}$ .) This again may be solved via the `scipy.stats.beta.ppf` function by rewriting the expression in a similar manner to (52). Solving for  $r_i$  is very similar; we begin by calculating:

$$R := \int_{p_i=1/k}^{p_i=1} \frac{(1-q)}{\binom{n}{x_1, x_2, \dots, x_k}} \binom{n}{x_i} (p_i)^{x_i} (1-p_i)^{n-x_i} dp_i \quad (55)$$

If  $R \geq A$ , then the value of  $r_i$  is the solution to:

$$\int_{p_i=r_i}^{p_i=1} \frac{(1-q)}{\binom{n}{x_1, x_2, \dots, x_k}} \binom{n}{x_i} (p_i)^{x_i} (1-p_i)^{n-x_i} dp_i = A, \quad (56)$$

which will be at least  $\frac{1}{k}$ . If  $R < A$ , we check if

$$R + \frac{q}{k^n} \geq A; \quad (57)$$

if it is, then  $r_i = \frac{1}{k}$ . If  $R + \frac{q}{k^n} < A$ , then  $r_i$  is the solution to:

$$\frac{q}{k^n} + \int_{p_i=r_i}^{p_i=1} \frac{(1-q)}{\binom{n}{x_1, x_2, \dots, x_k}} \binom{n}{x_i} (p_i)^{x_i} (1-p_i)^{n-x_i} dp_i = A \quad (58)$$

which will be at most  $\frac{1}{k}$ . These are the values of  $\ell_i$  and  $r_i$  provided by the program. The code may look strange for these credible intervals, as our algorithm is designed to minimize the risk of floating point errors.